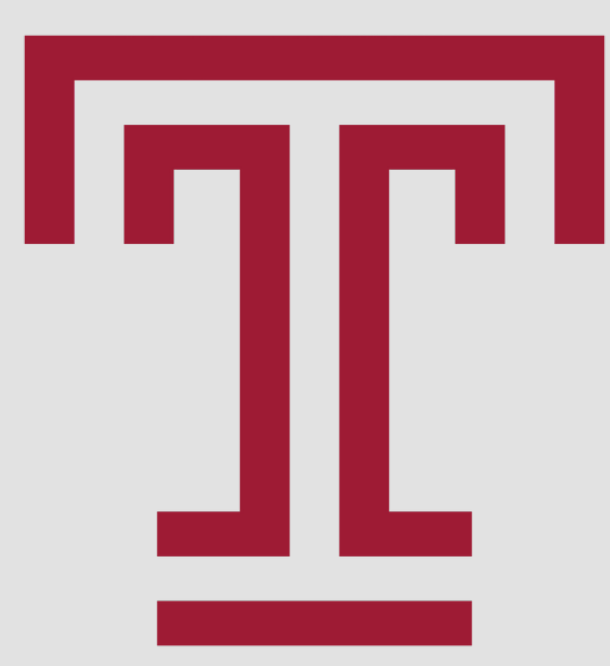
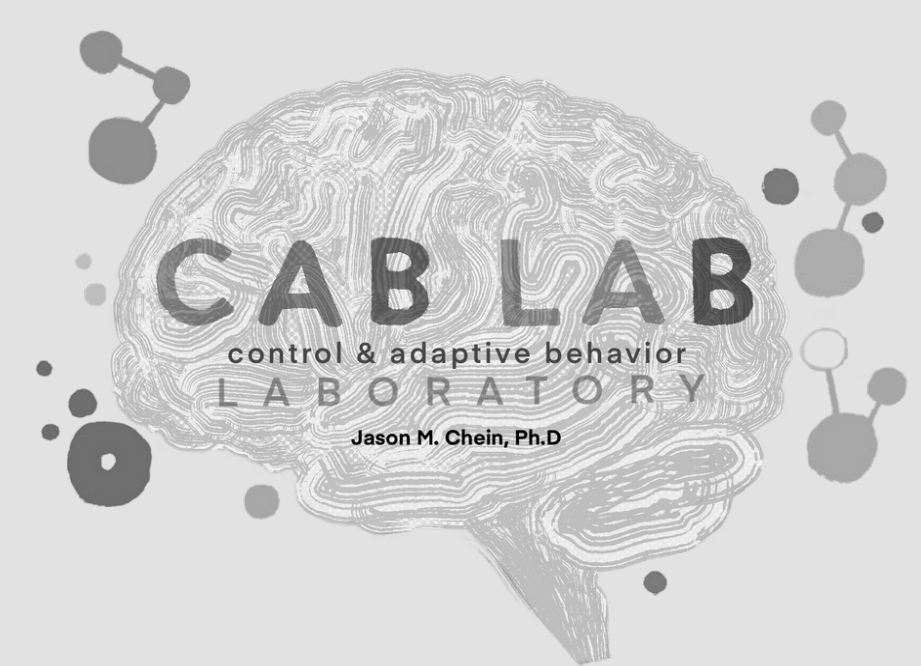




Children's Susceptibility to Content Generated by Artificial Intelligence

Allison Langer, Steven Martinez, Peter Marshall, Jason Chein
Department of Psychology and Neuroscience, Temple University



Background and Significance

- Advancements in artificial intelligence (AI) tools can make it difficult to discern what is “real”.
- While adults consistently perform near chance when discerning AI-generated texts and images (Dugan et al., 2023, Miller et al. 2023, Chein, Martinez, & Barone, 2024), no study has investigated this ability in children.
- Children are skeptical of implausible claims (Browne & Woolley, 2004) and untrustworthy informants (Sobel & Finiasz, 2020), however, the strategies they use to detect what is plausible and trustworthy may not translate to online environments (Shtulman, 2024).

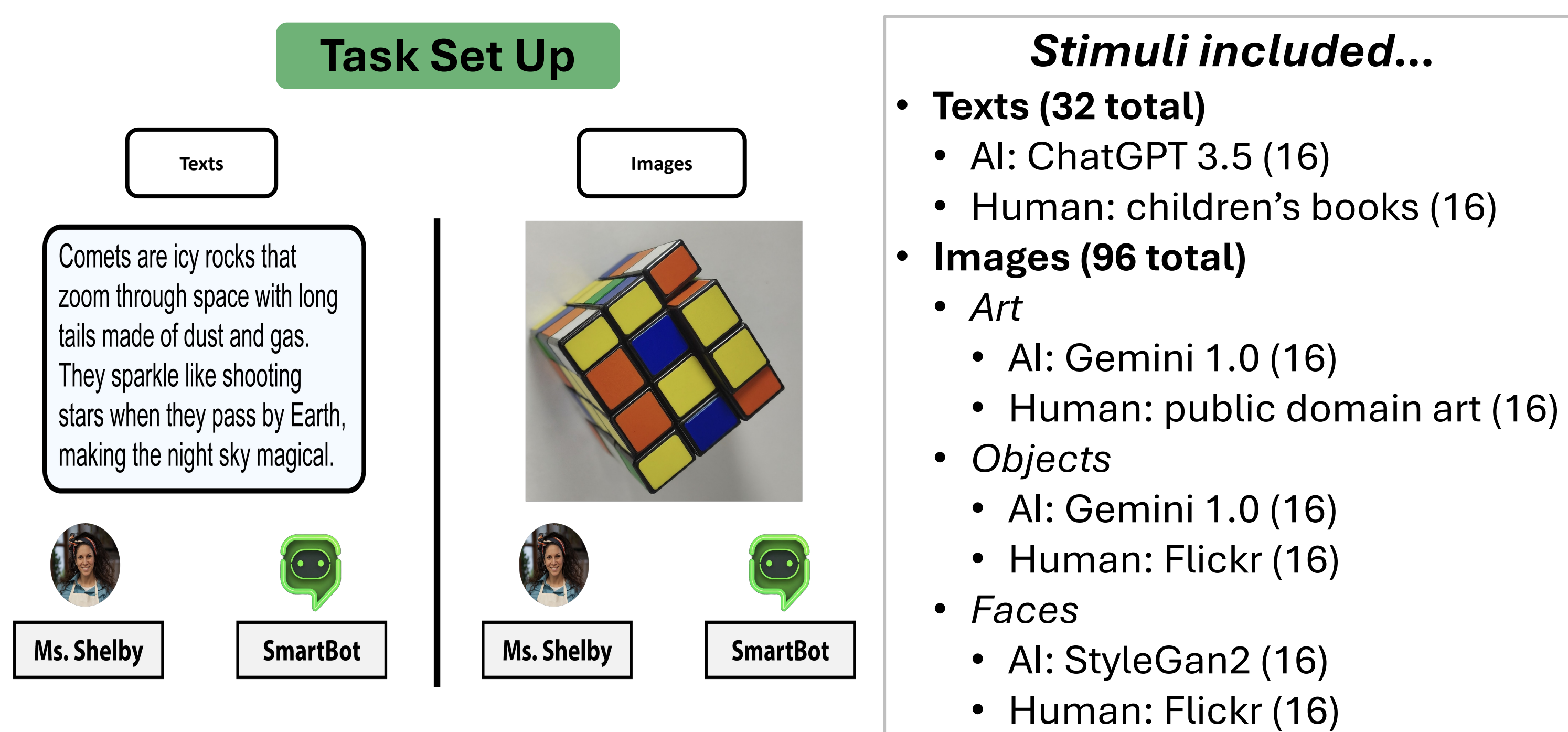
Here, we test if children can discern whether content is human- or AI- generated and examine if this ability is associated with fluid intelligence, receptive vocabulary, age, and technology exposure.

Methods

37 participants ages 6-10 years old (Mean=7.87, SD=1.11) completed the study via Zoom.

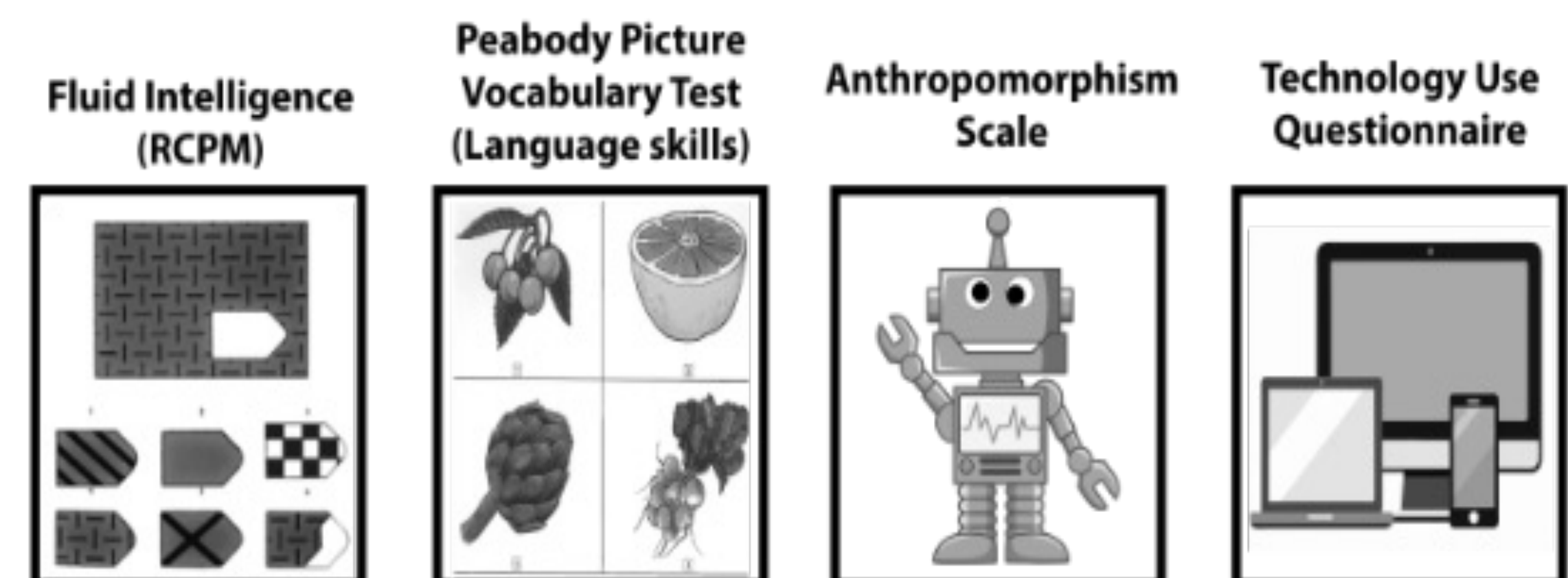
Human/AI Discernment Task

Children were provided with an age-appropriate narrative illustrating how a “SmartBot” (representing AI) and “Ms. Shelby” (representing the human), can write text, take photos, and create art. They were then asked to discern if stimuli came from the SmartBot or Ms. Shelby.



Psychological Characteristics

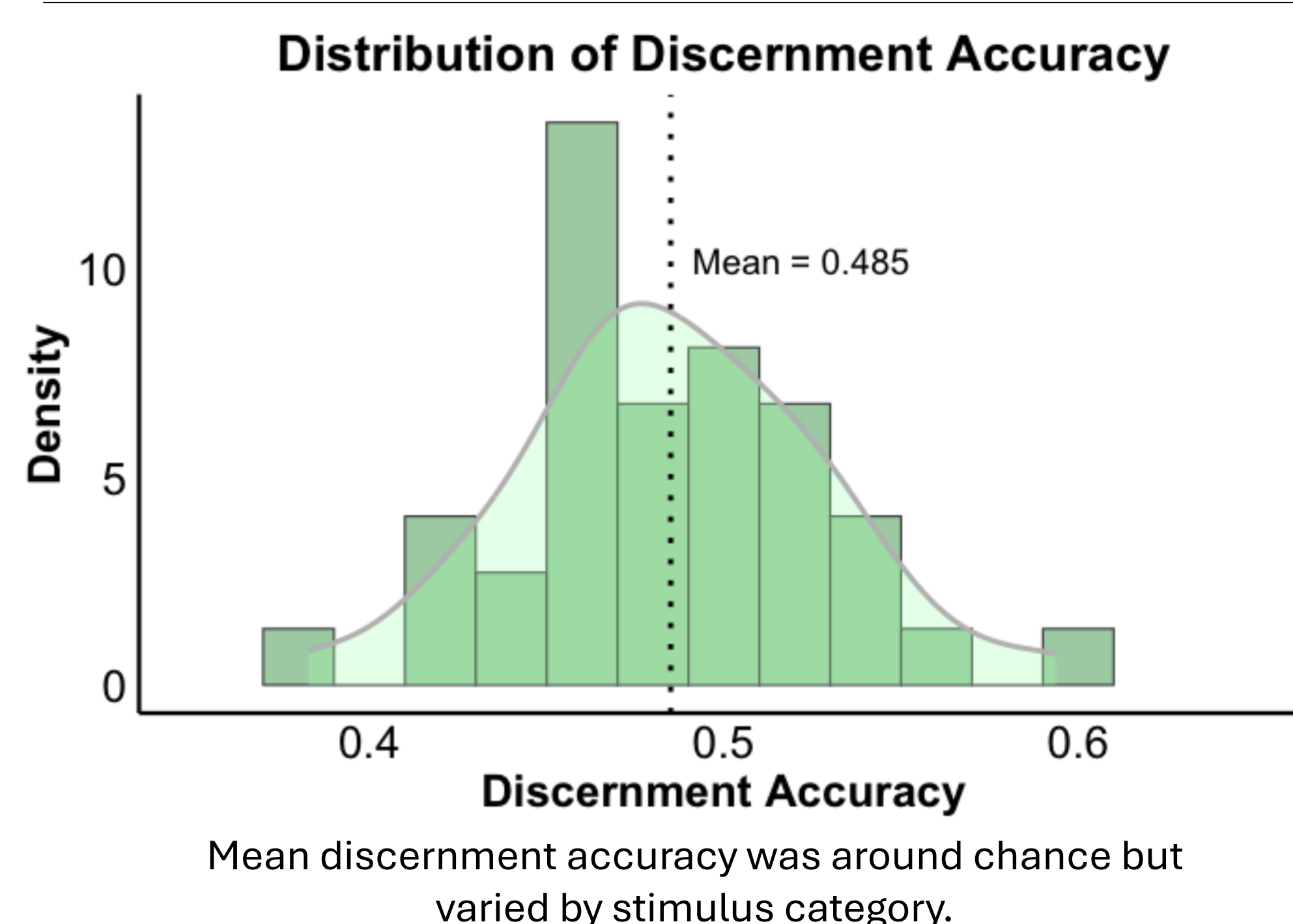
In addition to the human/AI judgment task, we collected individual difference measures:



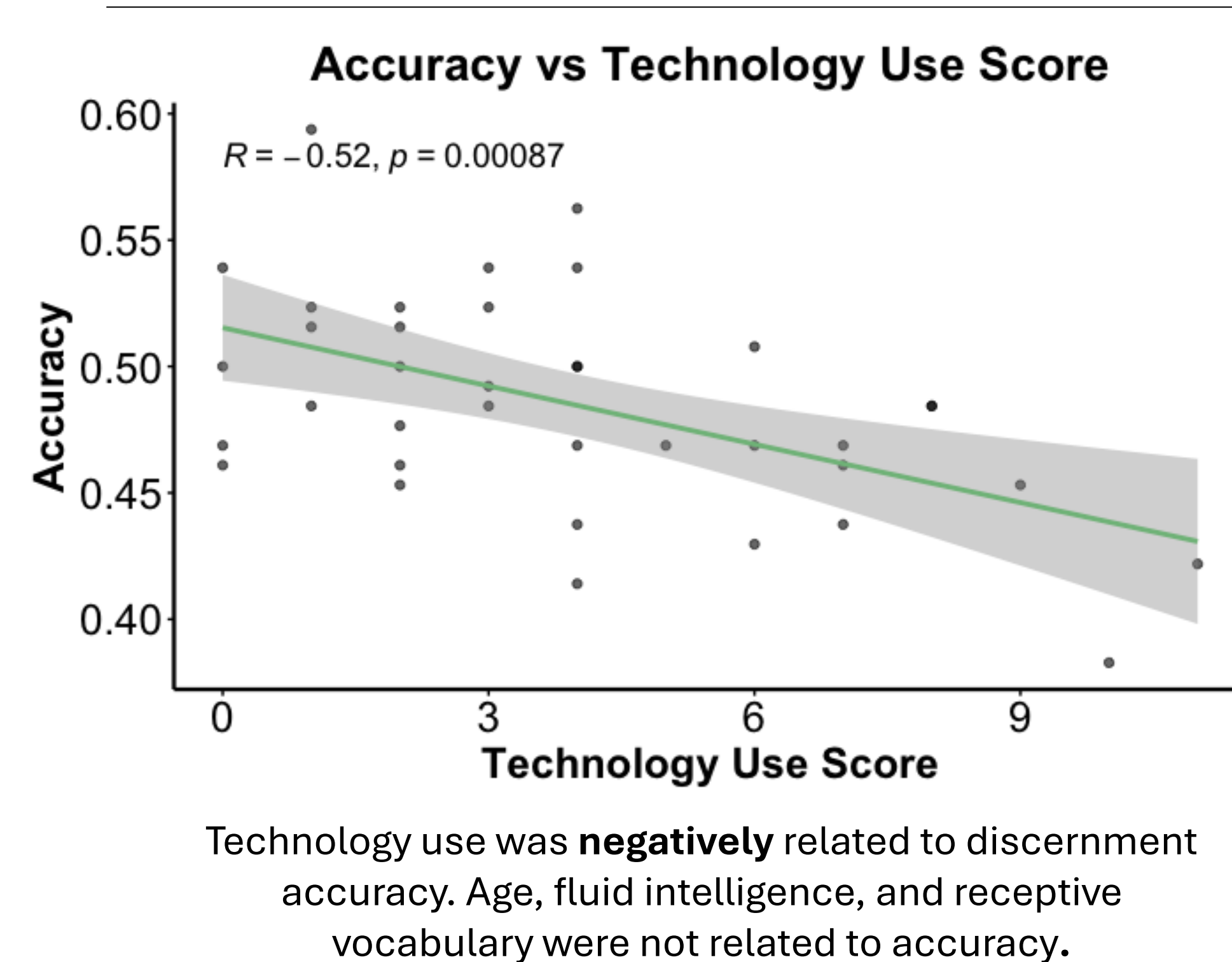
- 1)The Raven's Colored Progressive Matrices, as an index of nonverbal fluid intelligence
- 2)The Peabody Picture Vocabulary Test, as an index of language skills
- 3)An Anthropomorphism Scale to index the degree of human-like qualities children assign to the SmartBot
- 4)A parent-report technology use questionnaire measuring children's frequency of use of different devices like computers, iPads, & smartphones in the last week

Results

Discernment Accuracy

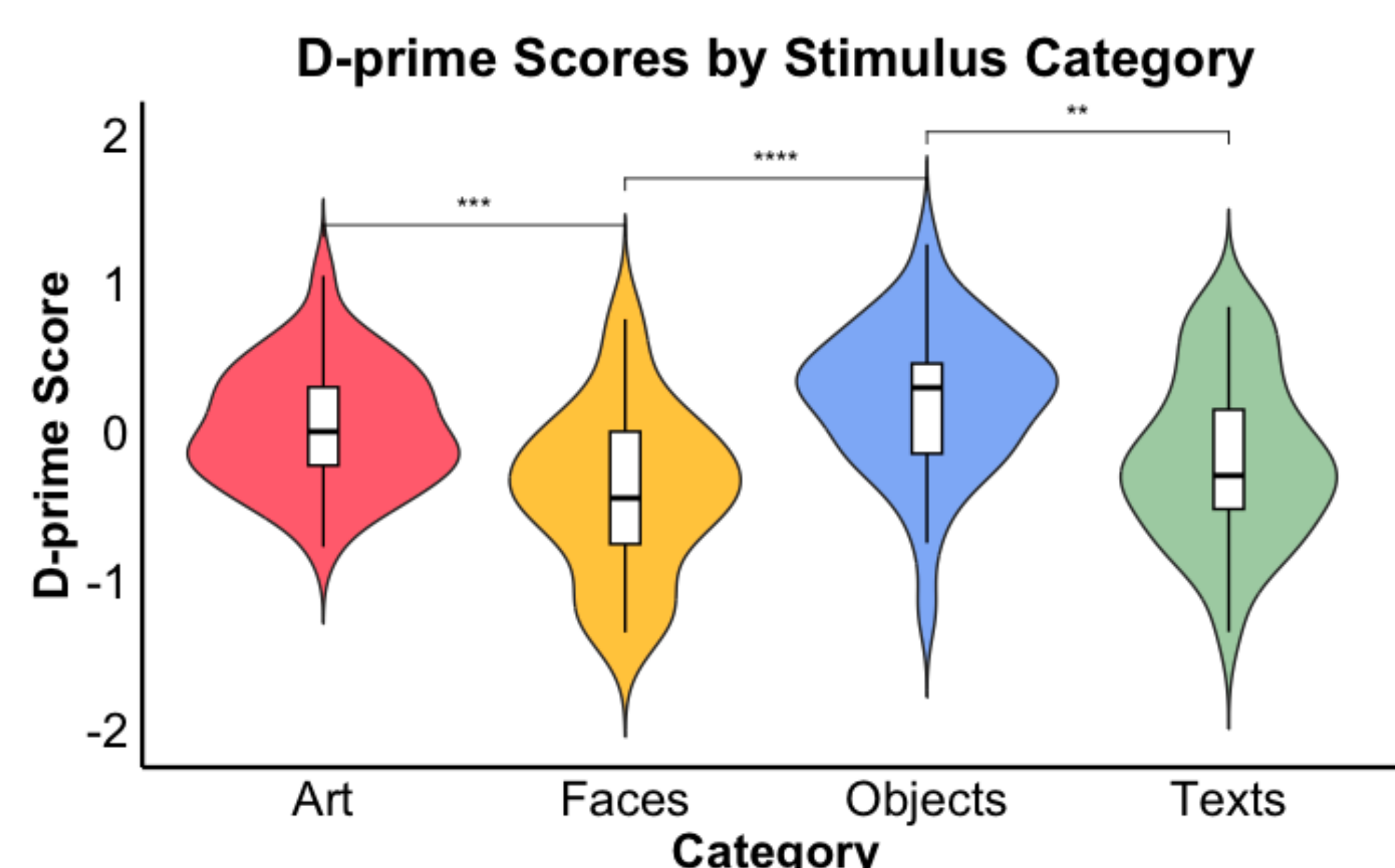
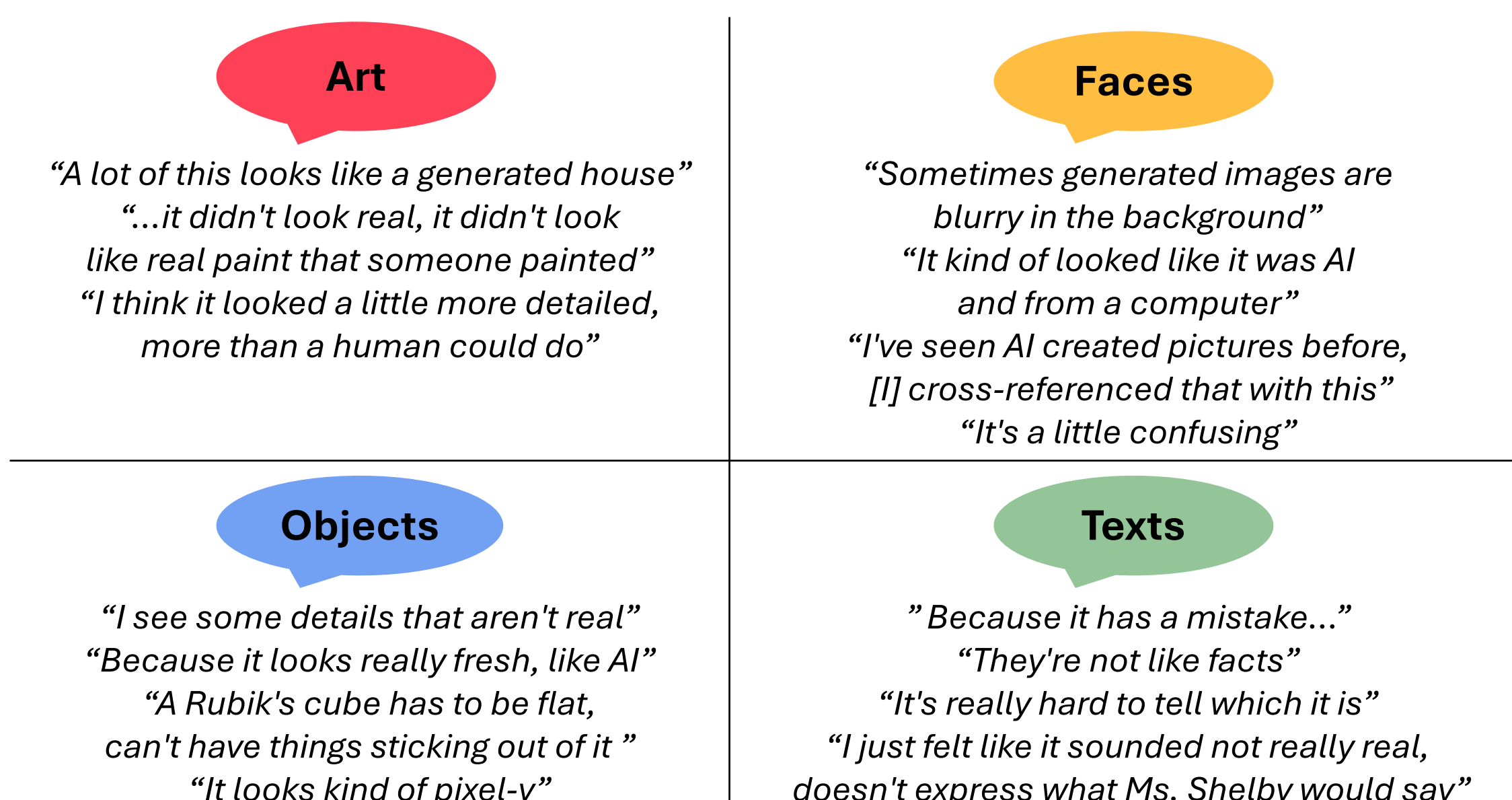


Predictors of Discernment Accuracy

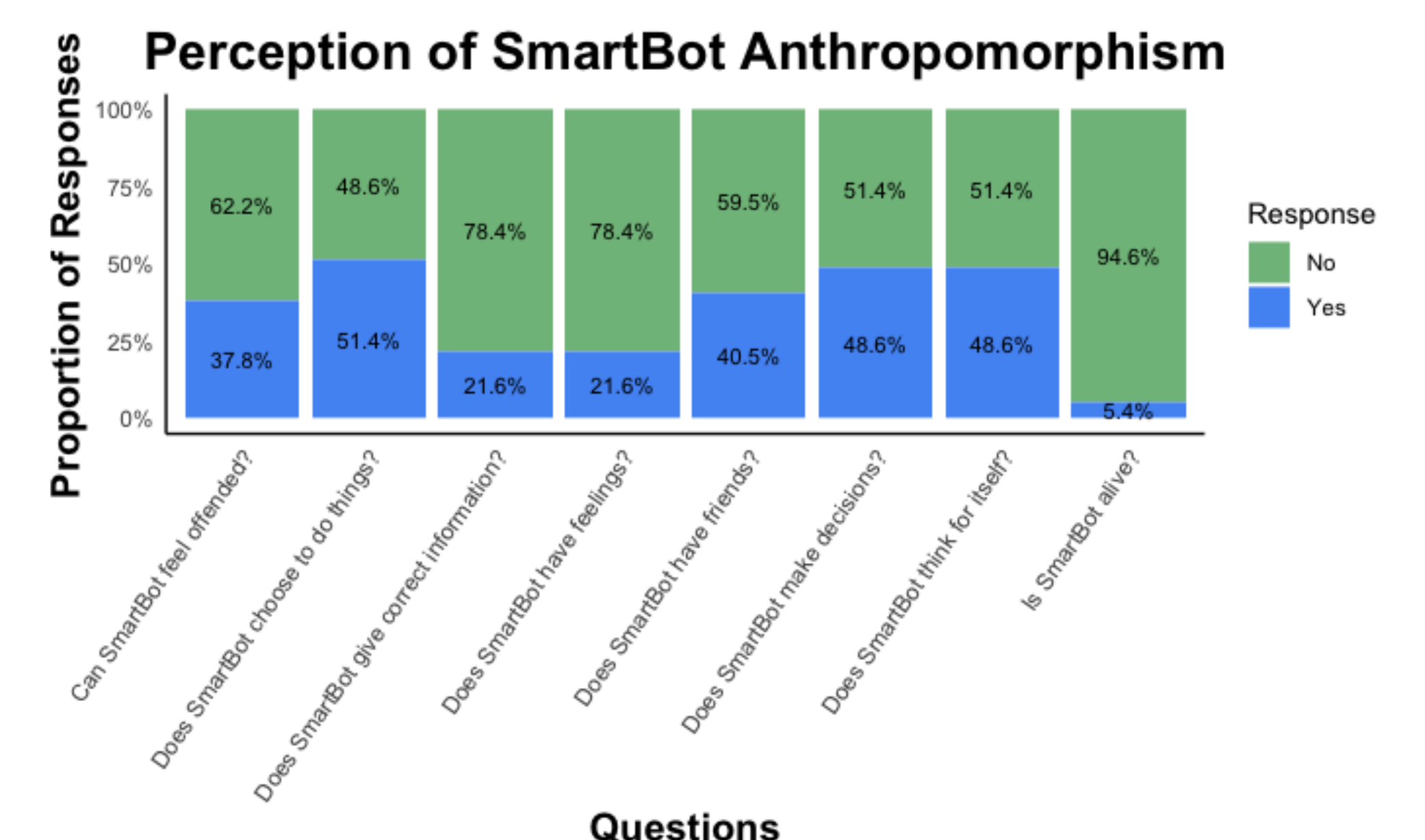
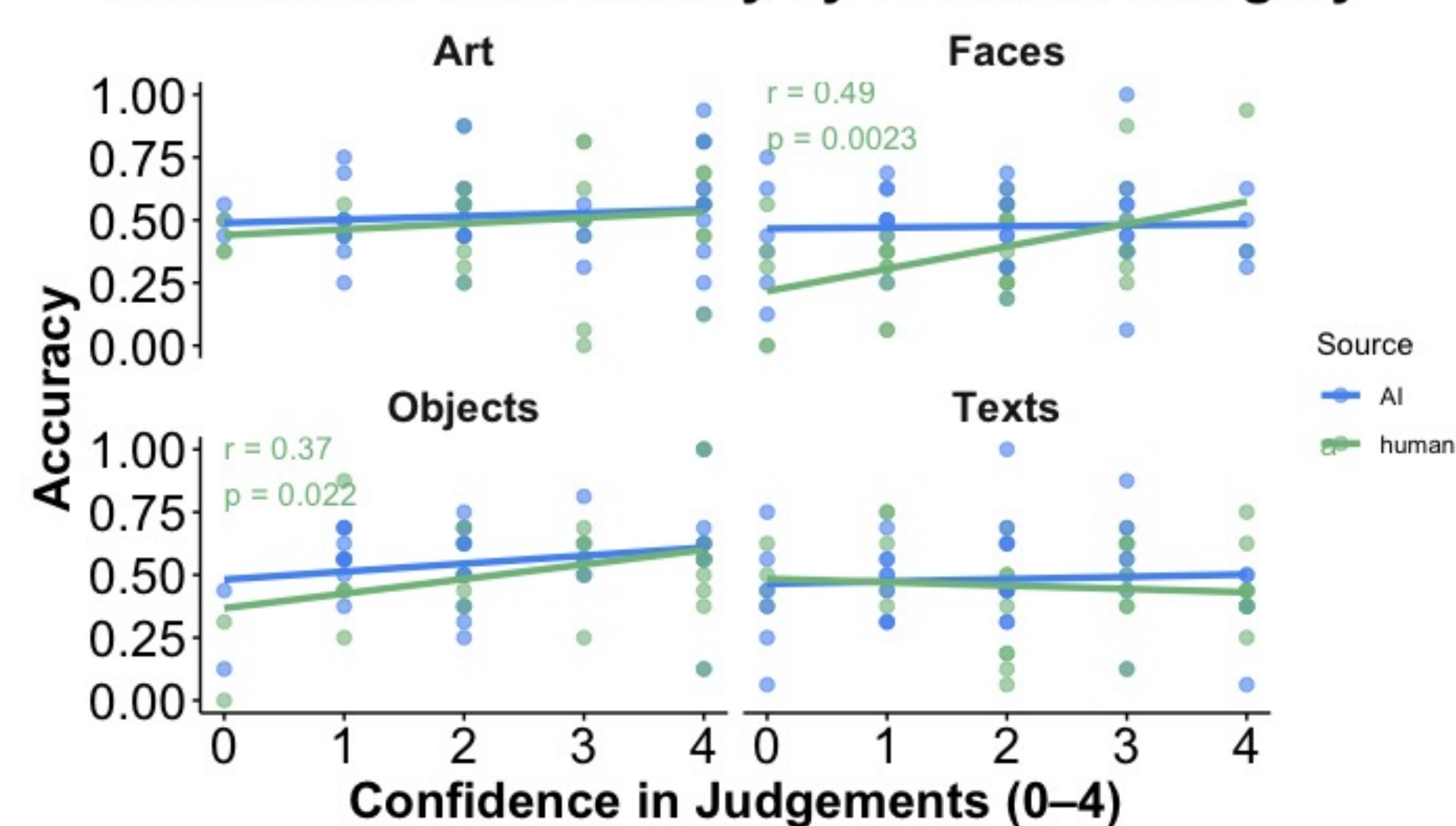


Perceptions of the SmartBot

“Can you tell me why you think this came from the SmartBot?”
Representative Participant Quotes



Confidence vs. Accuracy by Stimulus Category



Discussion

Though children used heuristics to make discernments, they were generally unable to detect AI-generated content from human-generated content. Detection accuracy varied by content type; objects and art were slightly easier to detect than faces and texts. Negative D' values suggest that children have a bias towards guessing content is **human-generated**. More frequent technology use was related to poorer discernment accuracy, suggesting spending more time on devices may lead to susceptibility to artificial content and risk for misattributing AI content as human. Our findings highlight the importance of early AI literacy to support children's engagement with AI technologies.